Mihnea-Ștefan Sândulache

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EDUCATION

University POLITEHNICA of Bucharest 2022 - 2026

Bachelor's Degree in Computer Science and Engineering *First Year Grade: 9.43/10*

- Relevant Coursework: **Operating Systems, Computer Programming, Data Structures and Algorithms, Numerical Methods, Object Oriented Programming, Algorithm Analysis**

EXPERIENCE

Machine Learning Engineer Trainee| *Raiffeisen Bank, Bucharest* August 2023 – February 2024

CSV Interpreter Chatbot | *Python*

- Developed a chatbot system that streamlines replies to user queries from CSV datasets using the OpenAI Azure API and a Langchain agent.
- Leveraged **prompt engineering** to enable the chatbot to understand user questions and generate contextually relevant responses.
- Generated different kind of plots, ensuring the chatbot's adaptability and enhancing user satisfaction through iterative improvement

Semantic Vault | *Python*

- Took part in the team that developed a **chatbot** which is trained to answer questions based on various documents divided in knowledge bases, each for a different department from the bank.
- Conducted **data analysis** by scrutinizing user interactions and inquiries with the chatbot, employing **Matplotlib** to craft insightful graphs and trends, providing a visually compelling representation of user engagement patterns.

Undergraduate Teaching Assistant| *University POLITEHNICA of Bucharest* October 2023 – Present

- Took part in the **Computer Programming team**, imparting essential skills in **C programming**, **Makefile** utilization, and fundamental **Linux** commands.
- Facilitated student learning by delivering **clear and concise explanations** of programming concepts, providing hands-on guidance through assignments and lab sessions, and **fostering logical problem-solving skills**.
- Contributed to curriculum development by proposing a **homework assignment** that encouraged students to apply their programming knowledge and improve their speech skills by assigning them to document it.

PROJECTS

Numerical Methods Project | *Matlab*

- **Markov chains:** Utilized a greedy heuristic approach to find the most probable path in the labyrinth, aiming to reach the WIN state, while using Markov chains to determine the path with the highest win probability.
- **Linear regression:** Created functions to load and split the MNIST dataset into training and testing data. Initialized weights for a neural network model and implemented the cost function.
- **Mnist 101:** Utilized efficient vectorization techniques to optimize computations and enhance performance. Implemented cost functions for Lasso and Ridge regression, thus expanding the scope of the regression analysis.

IMDB | *Java*

- Developed a Java Swing-based IMDB application with modular functionalities such as creating and deleting users, creating request ,which implies getting notifications, tailored for regular users, contributors, and administrators, featuring persistent data storage.
- Integrated design patterns including singleton, factory, observer, and strategy, enhancing code flexibility and maintaining an optimized user experience.

Sorting Algorithm Visualization | *C++*

- Built a **GUI Bubble Sort algorithm** which was implemented using **OOP concepts**, such as the vector class and its specific methods.
- Enhanced its performance using the **raylib library** which facilitated the building of all **graphical aspects**, as well as the logic and functionality of the application.

Image Compression Using Quadtrees| *C*

- Designed and implemented a powerful **command-line interface** (CLI) application for **image compression and decompression**, showcasing proficiency in **data structures and algorithms**.
- Leveraged advanced **quadtree data structures and algorithms** to efficiently compress and decompress images, optimizing the **representation of pixel data** and reducing PPM file sizes while preserving image quality.

Certificates and Awards

- **Cisco Certified Network Associate Module 1** at **Hackademy**
- **2nd Place at Olympiad of Society for Excellence and Performance in Informatics –** regional phase (2021)
- **2nd Place at National Olympiad of Informatics –** regional phase (2022)

Extracurricular Activity

- **Member** of the **Students' League**, performing activities at different events. (Oct 2022 – Present)
- **Mentoring the first year students**, thus **facilitating their transition** into university life and **empowering** them to navigate challenges with confidence, resulting in a **positive and lasting impact** on their academic and personal development.

Technical Skills

Programming Languages (Intermediate Level): **C**, **C++, Java**
Programming Languages (Basic Level): Python, Django, Vue.js, SQL, Assembly x86, Matlab
Programming Skills: Algorithms, Bash Scripting, Git, Linux, OOP, Data structures, Machine Learning, Design Patterns